

Statistical and Computational Models

Lecture 1: Statistical and computational foundations

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Department of Government, Cornell University · Fall 2026

Roadmap

The object of estimation

Estimating f

Loss and expected prediction error

What is 'Learning'?

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For example: Y_i is whether i voted, $\mathbf{X}_i$ is what we know about them, and $p(\mathbf{y} | \mathbf{x})$ tells us how the two are connected. We never observe $p(\mathbf{y} | \mathbf{x})$, instead we hope to learn it from a finite sample of pairs.

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Question: is linear regression a form of statistical learning? Yes! We 'learn' the $\hat{\beta}$ that minimises the sum of squared residuals.

Estimand

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With a coherent estimand in place, you can then turn to questions of **estimation**.

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Remember: the estimand is fixed and unknown. The estimate changes with every sample.

Some vocabulary

Statistics	Machine learning
estimation	learning, training
covariate, regressor, independent variable	feature
outcome, response, dependent variable	label, target
model selection	tuning, model selection
out-of-sample prediction error	generalization error, test error
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The same machinery is present in both literatures, but with very different vocabularies attached (helpful).

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It is worth being familiar with both; we will sometimes switch back and forth.

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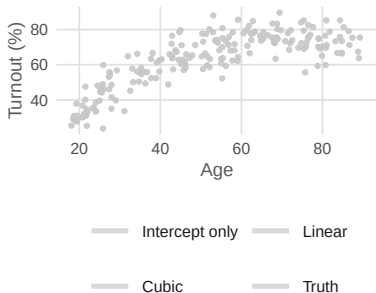
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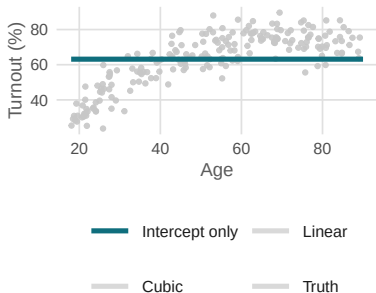
Change $\mathbf{X}$ and you change Y via $f(\cdot)$. This may or may not be causal.

Models as abstractions of reality



Suppose we have 200 age-by-state units (e.g. one might be the turnout rate for 20-year olds in NY).

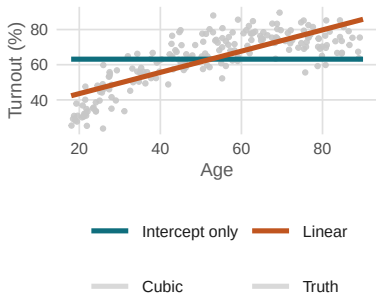
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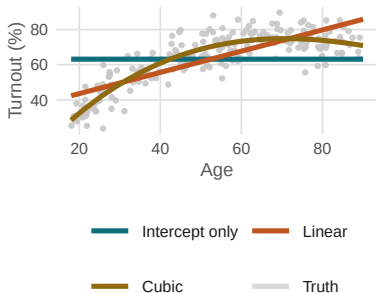


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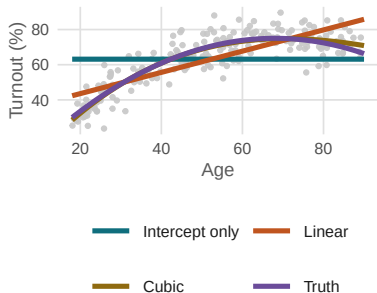
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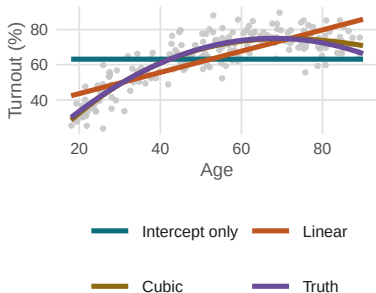
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Each fitted curve is a distinct **estimator** $\hat{f}$, returning its own **estimates** on a given draw.

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Read: the first term is a property of the **estimator** you chose. The second is a property of the **data generating process**.

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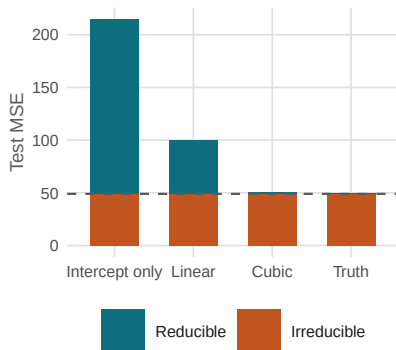
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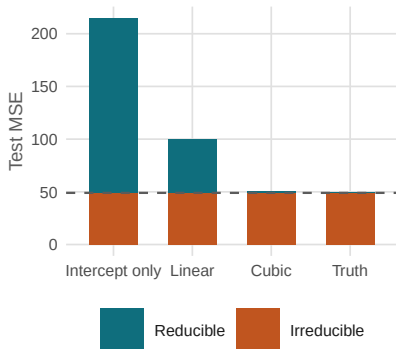
Remember, we can only **split** the last column because this is a simulation – with real data you see **test MSE alone**.

The error floor



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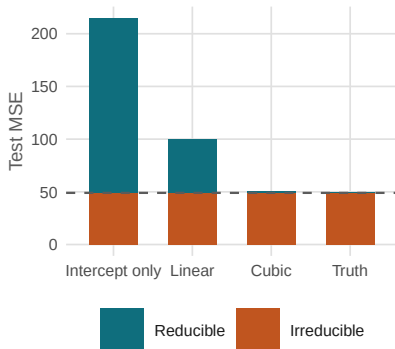
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Two routes to $\hat{f}$

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We will spend some time over the next two weeks thinking about what you trade off when choosing between the two.

Supervised and unsupervised

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'Supervised' because the outcome constrains the answer. It **teaches**.

Continuous Y is **regression**.

Categorical Y is **classification**.

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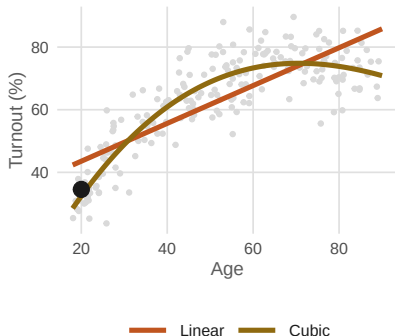
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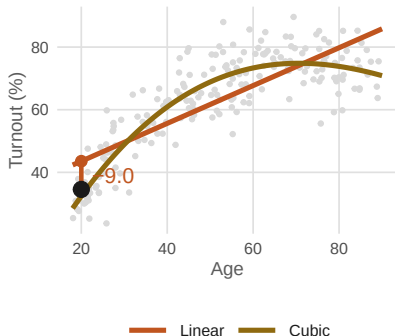
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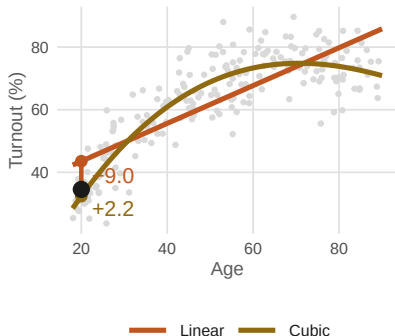
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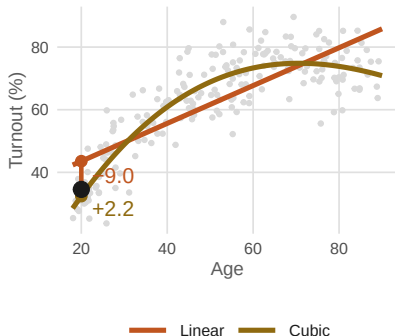


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For two different models the same unit gives us two different **errors**.

Two predictions on one case

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Linear	20	34.6	43.6	-9.0	81.0	9.0
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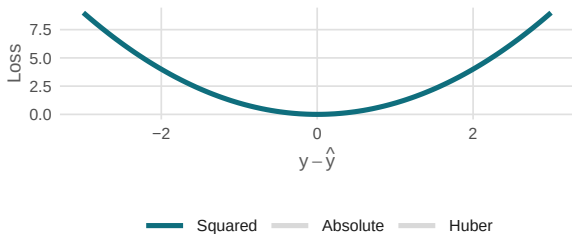
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So the two costs **need not agree**.

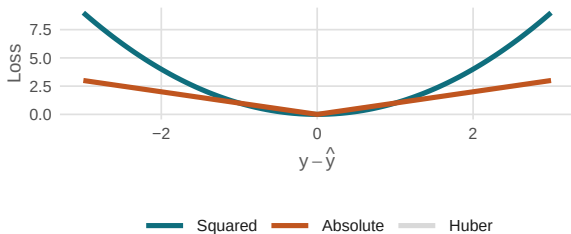
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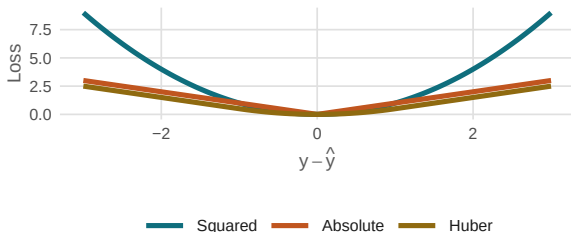


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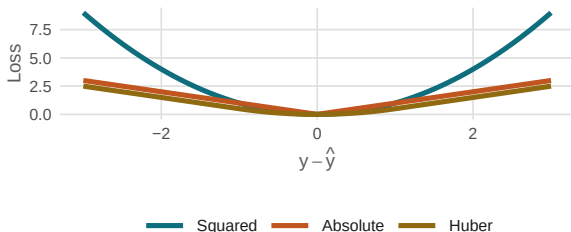
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Read: the loss function **prices the errors we necessarily make**. That choice is substantive, not a **computational convenience**.

Expected prediction error

$$\text{EPE}(f) = \mathbb{E} \left[L(Y, f(\mathbf{X})) \right] = \int L(y, f(\mathbf{x})) p(y, \mathbf{x}) dy d\mathbf{x}$$

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Note: ISL calls this the **expected test MSE** – squared loss, at one $\mathbf{x}$. Elsewhere you will meet it as the **risk**.

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Estimating EPE **in sample** causes problems, which we will address soon.

Training vs. test vs. EPE

Let's draw a very clear distinction early:

Quantity	Averages the loss over	Observable?
Training loss $\frac{1}{n} \sum_{i=1}^n L(y_i, \hat{f}(\mathbf{x}_i))$	the n observations we fit $\hat{f}$ on	always
Test loss $\frac{1}{m} \sum_{j=1}^m L(y_j, \hat{f}(\mathbf{x}_j))$	m observations not used in fitting $\hat{f}$	only if you have unseen data
EPE $\mathbb{E}[L(Y, \hat{f}(\mathbf{X}))]$	the population (or very many draws from the DGP)	never

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We will come back to this very soon!

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